

Dhruv Panchagatti

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U.S. Citizen • Eligible for U.S. Government Security Clearance

EDUCATION

University of California, Irvine — B.S. Electrical Engineering

Expected June 2030

Sophomore standing (60+ units via AP and Chapman University dual enrollment) • Focus: Digital Signal Processing & Embedded Systems

Northwood High School, Irvine, CA • GPA 4.36 (weighted)

June 2026

TECHNICAL SKILLS

Embedded & Hardware Microcontrollers (Arduino, ESP32), PWM motor control, circuit design, motor drivers, hardware interfacing, real-time control systems, soldering, CAD (Onshape, 3D printing)

Signal & Image Processing Hyperspectral imaging, spectral analysis, STFT/spectrogram feature extraction, MODTRAN atmospheric modeling, cross-sensor correlation

Programming Python, C++, Java

ML / Data CNNs, TensorFlow, regression modeling, feature selection

RESEARCH EXPERIENCE

Hyperspectral Imaging Researcher — UC Irvine Imaging & Computer Vision Lab (ICVL)

2025 – Present

Advisor: Prof. Glenn Healey • Remote sensing for early wildfire detection

- › Perform VNIR/SWIR pixel selection, alignment, and spectrum extraction on hyperspectral imagery to identify atmospheric signatures relevant to wildfire detection
- › Run MODTRAN-based atmospheric model fits against extracted pixel spectra, presenting matches for validation against defined error thresholds
- › Design and build an automated pipeline scaling the manual pixel-to-MODTRAN workflow, dramatically reducing per-pixel analysis time and enabling larger batch runs
- › Cross-reference VNIR and SWIR detector outputs against geo-referenced pixel locations to verify alignment and spectral consistency across bands

ENGINEERING PROJECTS

Electric Vehicle — autonomous distance-control system

1st Place, Science Olympiad Nationals

Designed and built a battery-powered vehicle from scratch; placed 1st of ~50 teams at the national tournament

- › Implemented closed-loop distance control on an Arduino using a rotary encoder for wheel-revolution counting, stopping the vehicle within 4 mm of the operator-set target across variable distances
- › Drove a 600 RPM DC motor through a motor driver with PWM ramping, tuning the acceleration and braking profile to trade top speed against overshoot
- › Built an LCD and rotary-encoder input interface for run-to-run target entry, and integrated a rifle-scope mount for repeatable start-line alignment

Arduino • Embedded C++ • Rotary encoder feedback • PWM motor control • LCD interface

Voice-Controlled Vehicle — embedded control system

Published, Scholarly Review Journal

- › Implemented PWM-based motor control and hardware interfacing across microcontrollers and motor drivers for precise real-time movement
- › Designed a two-stage CNN with STFT spectrogram feature extraction for real-time voice command recognition; built a custom command dataset and benchmarked architectures for accuracy
- › Integrated the trained model into a live robotic control system and architected the combined embedded hardware and software stack, with applications in assistive technology

Embedded C • Python • TensorFlow • PWM • Arduino

PUBLICATIONS

Panchagatti, D. "Voice-Controlled Vehicle: Determining the Best Machine Learning Models for Voice Recognition." Scholarly Review Journal. scholarlyreview.org/article/158298

Panchagatti, D. "Machine Learning in Soccer: Determining the Value of MLS Players Based on Performance Statistics." Scholarly Review Journal. scholarlyreview.org/article/143876

EXPERIENCE

AI Intern — GPT-Trainer / Petal / Go2Harvard • Remote

June – Sept 2024

- › Optimized AI system outputs through prompt engineering and retrieval-augmented generation; analyzed 500+ data points to improve performance

AWARDS & LEADERSHIP

Science Olympiad — Vice President; 1st Place, National Tournament (1st of ~50 teams, Electric Vehicle event); multiple state and regional engineering medals

California Science & Engineering Fair — 2nd Place, California State Tournament (300+ teams); State Qualifier, top 700 of 10,000+ projects; U.S. Army Award for excellence in research and innovation